

Miguel Otávio Cardoso Ramos

São José dos Campos, SP – Brazil | +55 12 98708-4762 | migueloct.ramos@gmail.com | [linkedin.com/in/miguelotavioramos](https://www.linkedin.com/in/miguelotavioramos) | github.com/miguelramoskskd | miguelramoskskd.github.io

SUMMARY

STEM / Computer Engineering student and researcher at Brazil's National Institute for Space Research (INPE), currently working on the research and development of software for lightning identification through image processing, combining scientific research, programming, and computer vision. My core interests include software development, artificial intelligence, computer vision, robotics, cloud computing, space science, linguistics and STEM education. My experiences in scientific research, robotics mentoring, and youth education have shown me that technology is most meaningful when it helps solve real-world problems and creates a positive impact on people's lives.

EDUCATION

B.S. in Science & Technology / Computer Engineering — UNIFESP — Institute of Science and Technology, São José dos Campos
2025 – 2029/2030 (expected) · GPA: 8.2/10

Technical High School Diploma in Business Administration — ETEC Prof. Ilza Nascimento Pintus
2023 – 2025

EXPERIENCE

Software Developer & Researcher

INPE — Brazil's National Institute for Space Research Undergraduate Fellowship in Technological Development and Innovation · Hybrid
Sep 2026 – Present

- Studying and applying the Phantom Cine Analyzer (PCA) documentation
- Analyzing brightness and other characteristics of scientific videos
- Developing methods to identify and temporally isolate lightning discharge events
- Building tools to process and export .cine files
- Automating steps of the lightning-video analysis workflow

Python · Computer Vision · YOLO/Ultralytics · Image Processing

PIBIC-EM Research Fellow

INPE — Brazil's National Institute for Space Research Supervised by Dr. Marcelo Magalhães Fares Saba · Hybrid
Sep 2025 – Jan 2026

- Analyzed lightning images recorded by specialized cameras using FLIR Tools and Phantom Camera Control
- Identified the challenges of analyzing large volumes of images without adequate hardware or a verification procedure
- Built a spreadsheet to organize files by priority, responsible researcher, location, verification status, and review date
- Collaborated on the verification workflow and documentation of the image database, allowing the research line to be continued by undergraduate students

Python · Excel · Research Projects · FLIR Tools · Data Organization

Volunteer Mentor — FIRST LEGO League Challenge

Quark 41 Team — Garage Team In-person
Sep 2023 – Jun 2026

- Provided technical and leadership mentorship to children and teenagers aged 10 to 15
- Contributed to the team's qualification for the FIRST LEGO League National Championship (2023/2024)
- Won the Core Values Award two years in a row, plus the Robot Design Award and the Coach/Mentor Excellence Award
- Developed evaluation methods to track each member's individual progress
- Contributed to the design, testing, and iteration of the competition robot, improving its reliability
- Applied project management methodologies: PDCA, Kanban, SCAMPER, and the Ishikawa Diagram

Arduino IDE · Python · C · TinkerCAD · Edge Impulse · ESP32-Cam · PDCA · Kanban

Clube Juvenil 14 Bis In-person

Activities Coordinator Jan 2026 – Present

- Coordinate workshops and sports, technical, educational, and recreational activities for children and teenagers

Activities Monitor Aug 2023 – Jan 2026

- Managed social media, data organization, and digital documentation

SKILLS

Programming Languages: Python, C

Web Development: HTML, CSS, JavaScript

AI/ML: YOLO/Ultralytics, Claude Code, Jupyter AI, Google Colab

Tools: Git, GitHub, VS Code, Arduino IDE, TinkerCAD, Microsoft Office

Languages: Portuguese (Native), English (Advanced), French (Intermediate), Spanish (Intermediate)

ACHIEVEMENTS

- Coach/Mentor Award — FIRST LEGO League, Regional Level, 2025/2026
- Robot Design Award — FIRST LEGO League, Regional Level, 2025/2026
- Core Values Award — FIRST LEGO League, Regional Level, 2024/2025 and 2023/2024
- Silver Medal — Brazilian Robotics Olympiad (OBR), Regional Level, 2024
- Bronze Medal — Brazilian Astronomy and Astronautics Olympiad (OBA), 2023